

Final Assignment: Credit Portfolio Risk

1. Assignment Description

Objective

Analyze the **one-year credit risk** of given portfolios using simulation. You will clean real-world-style messy data, construct a modeling dataset, estimate model inputs, simulate portfolio losses, and interpret and compare results.

Portfolios

You must analyze **Portfolio 1** and **Portfolio 2**. Analyzing **Portfolio 3** is optional (advanced).

Data

You are given multiple raw datasets — the data are intentionally **messy and incomplete**:

- `entities_info.csv` — firm characteristics
- `portfolio_*.csv` — portfolio holdings
- `stock_returns.csv` — firm returns
- `industry_index_returns.csv` — industry returns
- `default_history_20y.csv` — historical defaults

2. Tasks

Part 1 — Data Cleaning

Prepare a clean dataset suitable for modeling. At minimum:

- standardize **entity codes** and **industry labels**
- handle **missing values** (e.g., LGD, EAD, returns, defaults)
- remove or resolve **duplicates**
- ensure **consistent date formats**
- merge data into a usable modeling table

Part 2 — Model Setup and Parameter Estimation

Choose **one or two models** discussed in the course. For your chosen model(s):

- identify required inputs (e.g., PD, LGD, dependence structure)
- estimate parameters using the data

- justify assumptions or approximations

There is no single correct approach — clarity and reasoning are key.

Part 3 — Simulation and Risk Measurement

Simulate the **one-year loss distribution** and compute risk measures. You are free to choose the simulation design and level of complexity.

Part 4 — Analysis and Interpretation

Discuss your results. Possible directions:

- compare **Portfolio 1 vs Portfolio 2**
- compare results across models (if applicable)
- discuss the results
- (optional) analyze **Portfolio 3**

Focus on **clear reasoning, justified assumptions, and economic intuition**.

3. Deliverables

Code

Submit your Python code file(s) — there must be a single **main run file** that executes the full pipeline from start to finish. In your code, ensure:

- all outputs (plots, tables, cleaned data) are **clearly marked** with comments in the code indicating where they are saved and where they are used in the report
- the data directory is defined as a **single variable at the top**, e.g. `DIR_DATA = "/path/to/data"`, so it can be changed without modifying the rest of the code

Report

The report should be 3 to 5 pages in length, written in an academic style with clear structure (listed below). Use 12pt font (Times New Roman or similar serif font), 1.15 line spacing, and standard margins of 1 inch on all sides. Paragraphs should be separated by indentation. References should follow a consistent citation style (e.g. APA or Chicago).

The report must be structured as follows:

- **Data:** description of the dataset, cleaning procedures, and assumptions
- **Methodology:** model framework, parameter estimation, and relevant literature
- **Results and Discussion:** analysis and interpretation of results

Submission Instructions

Please submit your work as follows:

- Create a folder named `YourName_Final` in your branch containing:
 - your **code** file(s)
 - your **report** (PDF)
 - a **results/** subfolder with all outputs: cleaned data, plots, and tables
- As a contingency, **email the zipped folder** to `hongyi.shen@finance.uni-freiburg.de`

4. Grading Rubric

Grade Distribution

- Code: 30%
- Report: 60%
- Oral Exam: 10%

Oral Exam

The oral exam serves as a **verification of authorship and understanding**. Students are expected to explain their code, modeling choices, and results.

- If a student cannot demonstrate sufficient understanding, the assignment will be considered **failed**.
- The oral exam is not a standalone assessment, but a tool to ensure the work is **original and not fully AI-generated**.

General Requirements

- Work must be **original**. Students must be able to explain and justify their choices.
- **An AI-generated submissions will be considered a failure.**

1) Code

Essential:

- runs without errors
- clear structure with logical segmentation
- sufficient comments explaining key steps
- student demonstrates understanding of the implementation

Excellent:

- clean and efficient code structure
- use of reusable functions and modular design
- clear and well-organized documentation

2) Model and Analysis

Essential:

- at least one model is implemented correctly
- clear assumptions are stated and justified
- basic explanation of the model type
- appropriate risk metrics are computed (e.g., EL, VaR, ES)

Excellent:

- two models are implemented, justified, and compared
- strong understanding of the underlying statistical concepts
- multiple risk metrics are used and compared
- clear interpretation of differences across models and portfolios

3) Report (3–5 pages, main body)

Essential:

- written in an academic style (no AI-generated text)
- structured into: Data, Methodology, Results and Discussion
- length between 3 and 5 pages (excluding references, cover page, and table of contents)
- proper use of references and citation style

Excellent:

- clear and coherent academic writing
- sections are well integrated (not treated as isolated parts)
- demonstrates strong understanding of the statistical concepts
- results are clearly interpreted and linked to the methodology